Predicting Disease Outbreaks and Identifying the Factors Behind Case Surges

Project overview

This report presents the main results of the disease-outbreak project. It summarises the data, statistical tests, prediction models, important factors and district clusters. Tables and figures are loaded from results already produced by the project scripts.

The analysis describes reported outbreak records. It can identify patterns and associations, but it cannot prove that a factor causes an outbreak.

Project files

The report reads cleaned data from `data/processed/`, modelling data from `data/final/`, and saved tables and figures from `outputs/`. Run the relevant project scripts first if these files are missing.

Metric guide

Metric	Meaning
RMSE / MAE	Prediction error; lower is better.
R-squared	Variation explained by the regression model.
Precision / Recall	Precision reflects false alarms; recall reflects missed High outbreaks.
F1 / ROC-AUC / PR-AUC	Measures classification performance at or across thresholds.
Lift	Above 1 indicates a positive association in a rule.

The datasets

What the data files contain

`cleaned_epiclim.csv` holds one row per valid outbreak report after cleaning. `modeling_data.csv` adds the engineered features, the **split** (train, test or stress) and the two targets: `log_cases` for regression and `high_outbreak` for classification.

`Rows: 8970 | Columns: 35`

split	High	Low
stress	131	571
test	272	967
train	1750	5279

Key finding.

- The modelling dataset has **8,970** outbreak records and **35** columns.
- **stress:** 702 records, of which 131 (19%) are High outbreaks.
- **test:** 1,239 records, of which 272 (22%) are High outbreaks.
- **train:** 7,029 records, of which 1750 (25%) are High outbreaks.

The split is by calendar year, not at random. Models learn from 2009 to 2017, are judged on 2018 to 2019, and are checked once more on the sparse, COVID-era years 2020 to 2022. This mirrors real use, where a model predicts the future from the past.

Exploratory data analysis outputs

Outcome distribution

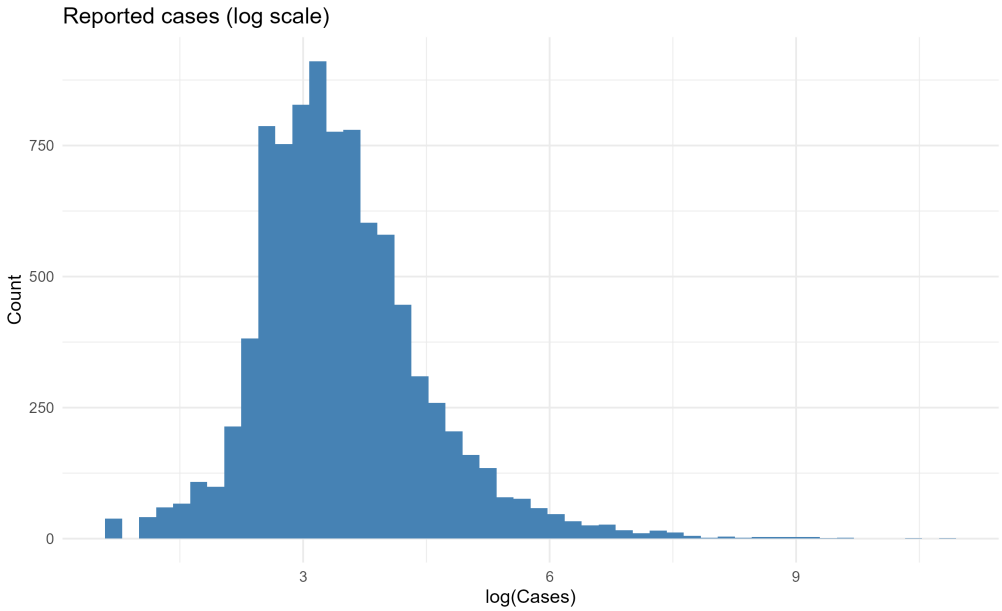


Figure 1: Figure 2: Reported cases on the log scale

Key finding.
Reported cases have a median of **27** and a mean of **89**; the largest record is **51,225** cases. A mean more than twice the median is the signature of a strongly right-skewed outcome, so the log scale is justified.

Disease burden

Disease	records	total_cases	median_cases	total_deaths
Acute Diarrhoeal Disease	5220	255796	28.0	1280
Dengue	1633	238335	23.0	621
Chikungunya	734	53343	25.0	14
Cholera	668	126520	35.0	461
Malaria	544	111117	35.0	564
Acute Encephalitis Syndrome	111	3848	9.0	1090
Dengue And Chikungunya	60	5580	25.5	6

`records` counts outbreak reports, `total_cases` sums reported cases, and `median_cases` shows the typical outbreak size. A disease with many records but a small median produces frequent small outbreaks; one with few records and a large median produces rare large ones.

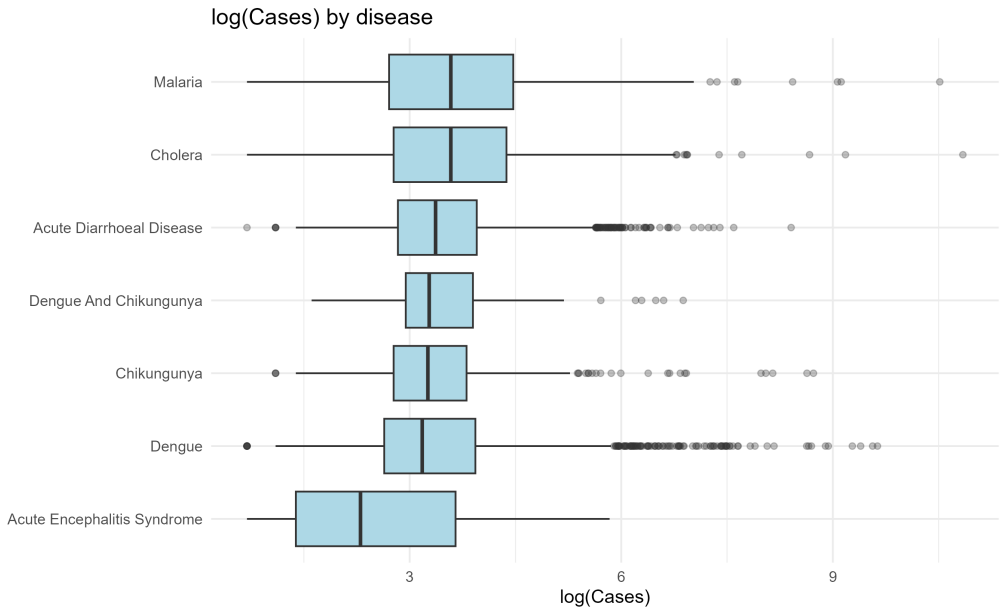


Figure 2: Figure 3: Log cases by disease

Key finding.
Acute Diarrhoeal Disease accounts for 58% of all records (5,220). The disease with the largest typical outbreak is **Cholera** (median 35 cases).

Time and seasonality

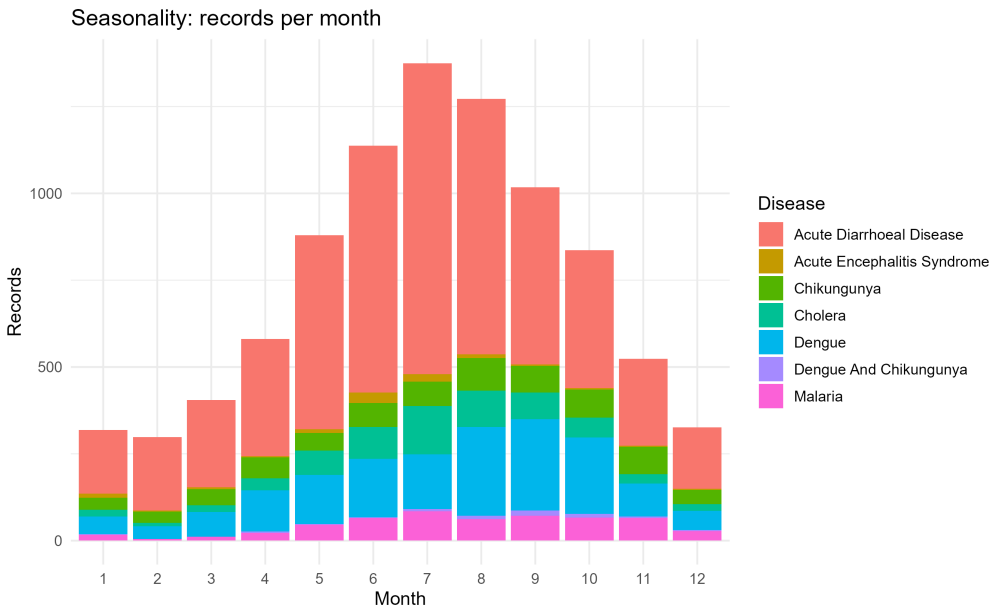


Figure 3: Figure 4: Outbreak reports per month, by disease

Key finding.
The busiest month for reports is **July** (15% of records), followed by **August** (14%).
Median log cases is highest in **Monsoon** (3.37) and lowest in **Winter** (3.22).

Geography

zone	records	mean_log_cases
Central	1271	3.48
East	1860	3.70
North	658	3.94
North-East	588	3.53
Other	20	2.79
South	2679	3.26
West	1894	3.37

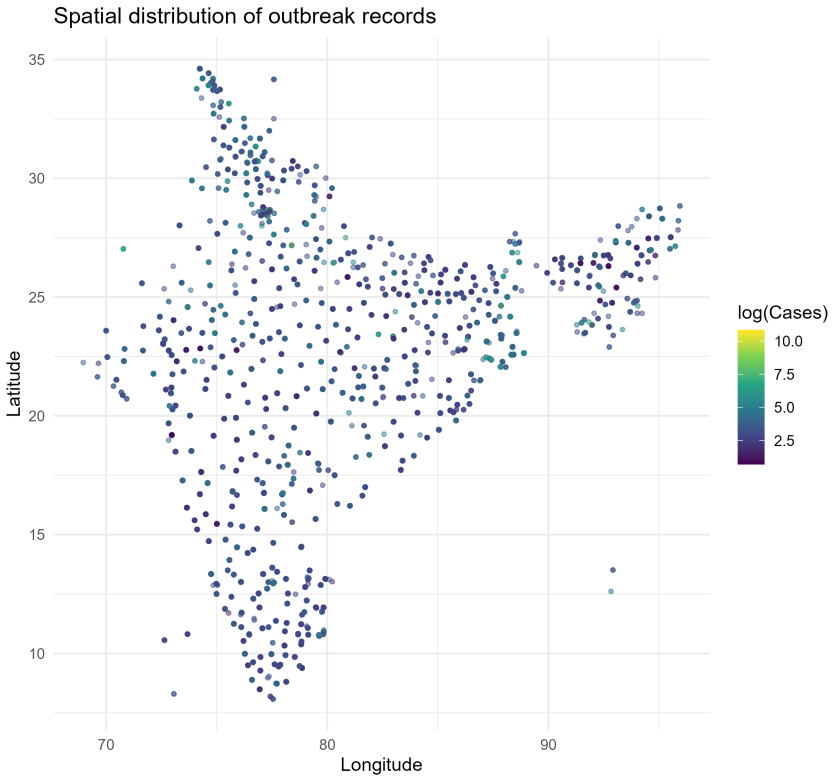


Figure 4: Figure 6: Spatial distribution of outbreak reports, coloured by log cases

Key finding.
Reports are most numerous in the **South** zone (2,679 records). Mean log cases is highest in **North** (3.94) and lowest in **Other** (2.79).

Climate relationships

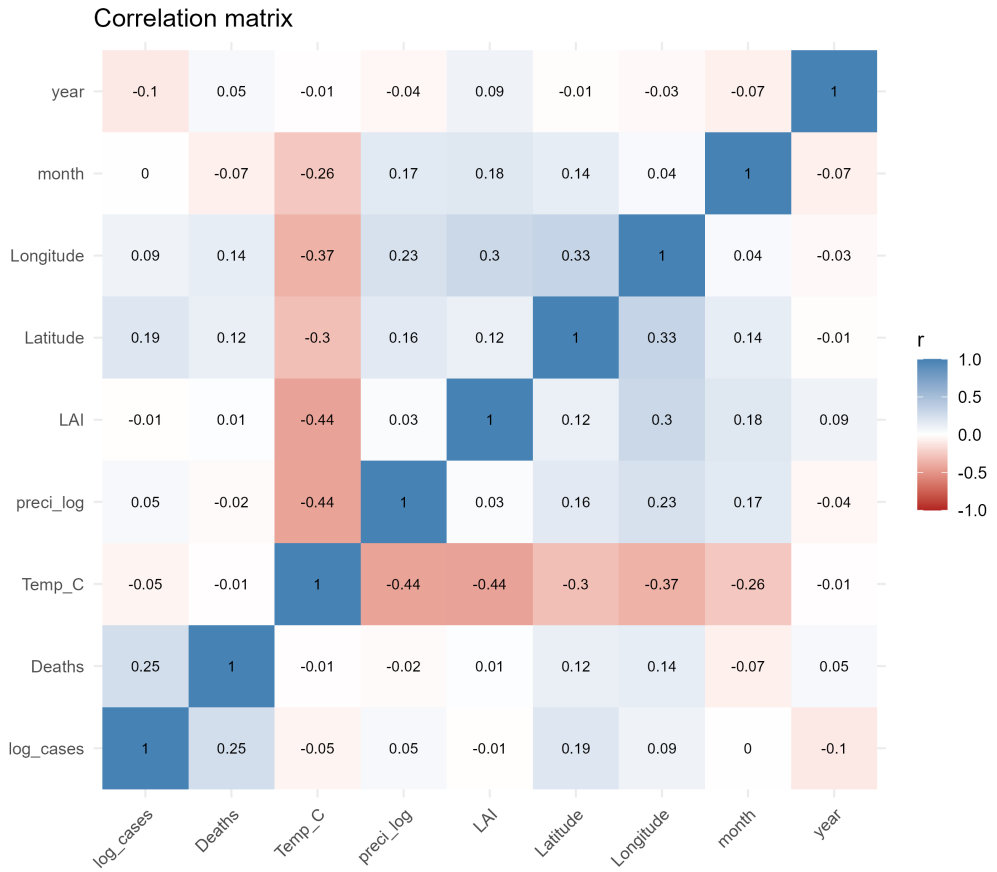


Figure 5: Figure 8: Correlation matrix

Key finding.

- **Temp_C**: correlation with log cases = -0.047 (negligible)
- **preci_log**: correlation with log cases = 0.050 (negligible)
- **LAI**: correlation with log cases = -0.007 (negligible)

All climate correlations are negligible. This is a legitimate finding: climate alone does not linearly track outbreak size here.

Missing data

Key finding.

- **Deaths**: 71.6% missing
- **LAI**: 24.4% missing
- **Temp_C**: 10.5% missing
- **preci**: 1.5% missing

Climate gaps are flagged and filled later using training-set medians only. Deaths is never filled and is never used as a predictor.

Statistical analysis outputs

These tables test whether the patterns seen in the figures are larger than chance. Records from the same district are not independent, so treat p-values as indicative and give more weight to effect sizes.

Group differences (Kruskal-Wallis)

variable	H	dof	p_value	epsilon_sq
Disease	69.8258	6	0.0000	0.0099
season	13.2392	3	0.0041	0.0019
zone	267.4747	5	0.0000	0.0381
state_ut	704.8146	35	0.0000	0.1003

`H` and `p_value` test whether log cases differ between groups. `epsilon_sq` is the effect size: roughly below 0.01 negligible, around 0.06 moderate, above 0.14 large. A tiny p-value with a tiny effect size means a real but practically unimportant difference.

Key finding.

- **Disease**: H = 69.8, p < 0.001, effect size 0.010 (negligible)
- **season**: H = 13.2, p = 0.004, effect size 0.002 (negligible)
- **zone**: H = 267.5, p < 0.001, effect size 0.038 (small)
- **state_ut**: H = 704.8, p < 0.001, effect size 0.100 (moderate)

Correlation with outbreak size (Spearman)

variable	n	rho	p_value	p_holm
roll3_log_cases	7029	0.2978	0.0000	0.0000
prev_log_cases	7029	0.2826	0.0000	0.0000
Latitude	7029	0.1847	0.0000	0.0000
days_since_prev	7029	-0.0653	0.0000	0.0000
Longitude	7029	0.0607	0.0000	0.0000
Temp_C	6256	-0.0484	0.0001	0.0009
month_cos	7029	-0.0448	0.0002	0.0010
preci_log	6924	0.0282	0.0190	0.0952
year_index	7029	-0.0256	0.0318	0.1272
LAI	5334	0.0169	0.2164	0.6492
month_sin	7029	-0.0103	0.3888	0.7777
n_prior_reports	7029	-0.0022	0.8530	0.8530

`rho` ranges from -1 to +1 and measures how consistently two variables rise or fall together without assuming a straight line. `p_holm` is the p-value adjusted for testing many variables at once.

Key finding.

- **roll3_log_cases**: rho = 0.30 (weak), n = 7,029, adjusted p < 0.001
- **prev_log_cases**: rho = 0.28 (weak), n = 7,029, adjusted p < 0.001
- **Latitude**: rho = 0.18 (weak), n = 7,029, adjusted p < 0.001
- **days_since_prev**: rho = -0.07 (negligible), n = 7,029, adjusted p < 0.001

Association with outbreak class and climate differences

variable	chi_sq	p_value	cramers_v
season	11.8057	0.0076	0.0410
zone	262.9815	0.0002	0.1934

variable	n	median_low	median_high	p_value
Temp_C	6256	30.653	29.590	0.000
preci_log	6924	0.199	0.263	0.000
LAI	5334	8.000	9.000	0.015

The chi-square table asks whether the share of High outbreaks differs between seasons or zones; `cramers_v` is the effect size. The climate table compares the median of each climate variable in Low and High outbreaks. Disease is excluded from the chi-square test because the High label is defined within each disease, which balances it by design.

Key finding.

- **season vs High outbreak**: p = 0.008, Cramer's V = 0.041 (negligible effect)
- **zone vs High outbreak**: p < 0.001, Cramer's V = 0.193 (small effect)
- **Temp_C**: median 30.65 in Low vs 29.59 in High outbreaks (p < 0.001)
- **preci_log**: median 0.20 in Low vs 0.26 in High outbreaks (p < 0.001)
- **LAI**: median 8.00 in Low vs 9.00 in High outbreaks (p = 0.015)

Multicollinearity (VIF)

variable	vif
preci_log	25.77
LAI	25.59
temp_x_precip	22.76
temp_x_lai	22.00
roll3_log_cases	4.02
prev_log_cases	3.98
Temp_C	3.63
month_cos	1.51
Longitude	1.39
month_sin	1.38
Latitude	1.35
n_prior_reports	1.27
year_index	1.18
days_since_prev	1.14

A variance inflation factor above 5 to 10 means a predictor is largely explained by the others, which makes linear-model coefficients unstable. Interaction terms usually score high against their own components.

Key finding.

Variables with VIF above 5: preci_log (25.8), LAI (25.6), temp_x_precip (22.8), temp_x_lai (22). Interpret their linear-regression coefficients with caution.

Regression outputs: predicting outbreak size

The regression models predict `log_cases`. They are always compared with a **baseline** that predicts each disease's typical (median) size. A model is only useful if it clearly beats that baseline.

set	model	n	RMSE	MAE	R2	MAE_cases
stress	Baseline	702	1.116	0.881	-0.020	42.384
stress	Linear Regression	702	1.040	0.813	0.115	40.467
stress	Random Forest	702	1.032	0.807	0.129	38.559
test	Baseline	1239	1.050	0.790	0.004	72.344
test	Linear Regression	1239	0.961	0.740	0.166	69.640
test	Random Forest	1239	0.951	0.738	0.183	69.992

Each row is a model on one evaluation set. Compare each model's RMSE and R-squared with the `Baseline` row for the same set. `MAE_cases` converts the error back to real cases but is dominated by a few huge outbreaks, so rely on the log-scale columns.

Key finding.

- **Test years (2018 to 2019):** the best model is **Random Forest** with RMSE 0.95 against 1.05 for the baseline, which is a modest improvement over the baseline (9.4% change in error). R-squared is 0.18, which is modest.
- **Stress years (2020 to 2022):** the best model is **Random Forest** with RMSE 1.03 against 1.12 for the baseline, which is a modest improvement over the baseline (7.6% change in error). R-squared is 0.13, which is modest.

Performance by outbreak history

set	model	has_history	n	RMSE	MAE	R2	MAE_cases
stress	Baseline	0	66	1.402	1.075	0.109	137.424
stress	Baseline	1	636	1.083	0.861	-0.055	32.521
stress	Linear Regression	0	66	1.393	1.072	0.119	137.534
stress	Linear Regression	1	636	0.996	0.786	0.107	30.394
stress	Random Forest	0	66	1.413	1.068	0.094	138.843
stress	Random Forest	1	636	0.984	0.780	0.129	28.152
test	Baseline	0	158	1.209	0.891	0.034	265.642
test	Baseline	1	1081	1.025	0.775	-0.008	44.092
test	Linear Regression	0	158	1.251	0.947	-0.035	267.473
test	Linear Regression	1	1081	0.911	0.710	0.205	40.724
test	Random Forest	0	158	1.204	0.904	0.042	265.247
test	Random Forest	1	1081	0.908	0.714	0.209	41.453

`has_history = 1` rows belong to a district and disease with an earlier report, so history features carry real information. `has_history = 0` rows only receive a disease-median stand-in, so models should perform much like the baseline there.

Key finding.

- **Linear Regression**, no earlier report: RMSE 1.25, R-squared -0.04 (n = 158)
- **Linear Regression**, with an earlier report: RMSE 0.91, R-squared 0.20 (n = 1,081)
- **Random Forest**, no earlier report: RMSE 1.20, R-squared 0.04 (n = 158)
- **Random Forest**, with an earlier report: RMSE 0.91, R-squared 0.21 (n = 1,081)

Figures

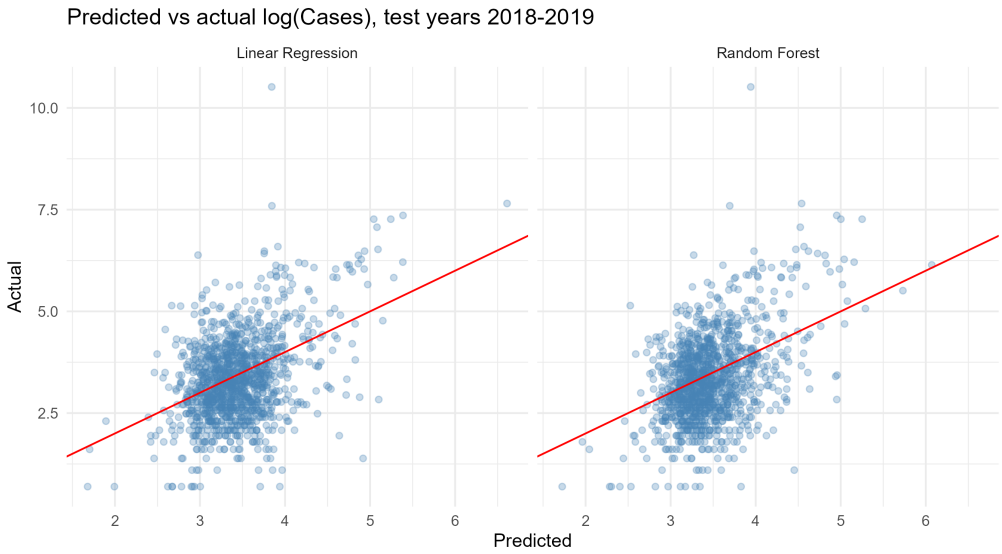


Figure 6: Figure 10: Predicted against actual log cases, test years

Each dot is one outbreak. Dots hugging the red diagonal are well predicted. A flat horizontal band of dots means the model predicts nearly the same value for everything and has little skill. Dots far above the line are outbreaks the model underestimated.

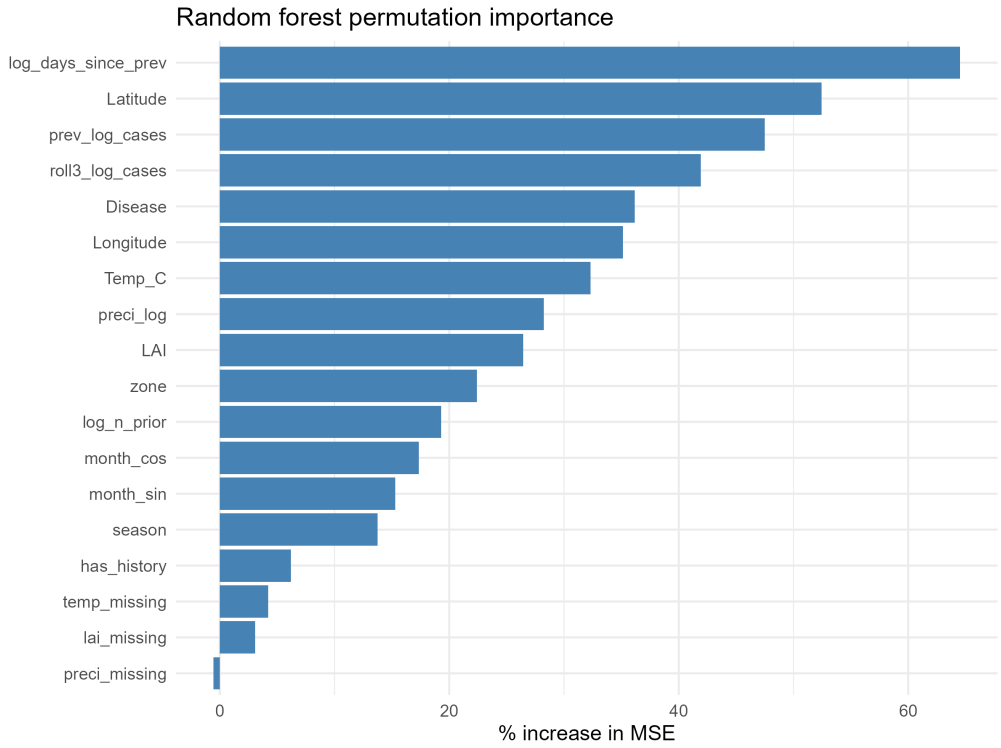


Figure 7: Figure 12: Random forest permutation importance (regression)

Figure 12 shows how much prediction error increases when a variable is scrambled: longer bars mean the model relies on that variable more. Figure 13 compares error distributions: lower boxes mean smaller errors, and the baseline box shows what a trivial rule achieves.

Coefficients and importance

term	Estimate	Std. Error	t value	Pr(>|t|)	2.5 %	97.5 %
(Intercept)	1.180	0.489	2.413	0.016	0.222	2.139
DiseaseAcute Encephalitis Syndrome	-0.520	0.101	-5.123	0.000	-0.719	-0.321
DiseaseChikungunya	0.160	0.046	3.503	0.000	0.070	0.249
DiseaseCholera	0.114	0.042	2.680	0.007	0.031	0.197
DiseaseDengue	0.147	0.033	4.475	0.000	0.082	0.211
DiseaseDengue And Chikungunya	0.474	0.136	3.488	0.000	0.207	0.740
DiseaseMalaria	0.189	0.047	3.996	0.000	0.096	0.281
seasonPost-monsoon	0.065	0.058	1.131	0.258	-0.048	0.179
seasonSummer	-0.010	0.062	-0.170	0.865	-0.131	0.110
seasonWinter	0.090	0.094	0.955	0.339	-0.095	0.275
zoneEast	0.092	0.055	1.658	0.097	-0.017	0.200
zoneNorth	0.251	0.070	3.595	0.000	0.114	0.387
zoneNorth-East	0.011	0.099	0.115	0.908	-0.182	0.205
zoneSouth	0.016	0.061	0.259	0.796	-0.104	0.136
zoneWest	0.045	0.050	0.902	0.367	-0.053	0.143
Latitude	0.020	0.005	3.921	0.000	0.010	0.029
Longitude	0.008	0.006	1.320	0.187	-0.004	0.019
Temp_C	0.003	0.002	1.200	0.230	-0.002	0.008
preci_log	0.053	0.045	1.175	0.240	-0.035	0.142
LAI	-0.001	0.002	-0.264	0.792	-0.005	0.003
month_sin	0.036	0.042	0.863	0.388	-0.046	0.119
month_cos	-0.079	0.041	-1.904	0.057	-0.160	0.002
prev_log_cases	0.205	0.025	8.341	0.000	0.157	0.254
roll3_log_cases	0.215	0.028	7.587	0.000	0.160	0.271

term	Estimate	Std. Error	t value	Pr(>|t|)	2.5 %	97.5 %
log_days_since_prev	-0.053	0.006	-8.663	0.000	-0.065	-0.041
log_n_prior	0.004	0.016	0.276	0.783	-0.027	0.036
has_history	-0.184	0.039	-4.723	0.000	-0.261	-0.108
preci_missing	-0.128	0.094	-1.365	0.172	-0.312	0.056
lai_missing	-0.026	0.026	-1.003	0.316	-0.077	0.025
temp_missing	-0.031	0.040	-0.773	0.440	-0.108	0.047

Key finding.
In the linear model, **12** terms are statistically significant at the 5% level (indicative only, because records are not independent).

- **log_days_since_prev**: coefficient -0.053 ($p < 0.001$)
- **prev_log_cases**: coefficient 0.205 ($p < 0.001$)
- **roll3_log_cases**: coefficient 0.215 ($p < 0.001$)
- **DiseaseAcute Encephalitis Syndrome**: coefficient -0.520 ($p < 0.001$)
- **has_history**: coefficient -0.184 ($p < 0.001$)

The random forest relies most on: **log_days_since_prev, Latitude, prev_log_cases, roll3_log_cases, Disease**.
In a linear model on log cases, a coefficient of 0.1 on a variable means one extra unit of it goes with roughly a 10% change in cases, holding the others fixed. Category coefficients are relative to the first (reference) category alphabetically.

Classification outputs: predicting surges

A report is a **High outbreak** when its cases exceed the 75th percentile of its own disease, calculated on training years.

Disease	n_train	outbreak_threshold
Acute Diarrhoeal Disease	4172	51.0
Acute Encephalitis Syndrome	90	43.0
Chikungunya	517	48.0
Cholera	572	78.2
Dengue	1168	56.0
Dengue And Chikungunya	48	50.5
Malaria	462	96.8

`outbreak_threshold` is the case count above which a report is High for that disease. Small `n_train` values mean the threshold rests on few records and is less reliable.

set	model	n	threshold	accuracy	precision	recall	F1	specificity	ROC_AUC	PR_AUC	prevalence
test	Logistic Regression	1239	0.22	0.639	0.323	0.585	0.416	0.655	0.670	0.444	0.220
test	Random Forest	1239	0.27	0.670	0.341	0.540	0.418	0.706	0.684	0.426	0.220
test	Previous report rule	1239	0.50	0.738	0.399	0.379	0.389	0.840	0.609	0.318	0.220
test	Always Low	1239	0.50	0.780	NA	0.000	0.000	1.000	0.500	0.220	0.220
stress	Logistic Regression	702	0.22	0.695	0.307	0.504	0.382	0.739	0.665	0.357	0.187
stress	Random Forest	702	0.27	0.688	0.314	0.565	0.403	0.716	0.670	0.382	0.187
stress	Previous report rule	702	0.50	0.756	0.341	0.328	0.335	0.855	0.591	0.248	0.187
stress	Always Low	702	0.50	0.813	NA	0.000	0.000	1.000	0.500	0.187	0.187

Compare each model with two baselines. *Always Low* shows how misleading accuracy is. *Previous report rule* predicts High if the last report in that district and disease was above the threshold; a useful model has to beat it. Compare PR-AUC with `prevalence`, the no-skill level.

Key finding.

- On the test years, **22%** of outbreaks are High (the no-skill PR-AUC level).
- The best model is **Logistic Regression**: PR-AUC 0.44 (2.0 times the no-skill level), ROC-AUC 0.67.
- At its chosen threshold of 0.22 it reaches precision 0.32, recall 0.58 and F1 0.42.
- It beats the simple previous-report rule (PR-AUC 0.44 vs 0.32) by 0.13.
- A model that always predicts Low is right 78% of the time yet catches no High outbreaks, which is why accuracy alone is misleading.

Confusion matrix

`Best model: Logistic Regression | threshold: 0.22`

	Low	High
Low	633	334
High	113	159

Rows are the truth and columns are the prediction. The bottom-right cell holds correct High predictions, the top-right holds false alarms, and the bottom-left holds High outbreaks the model missed. Precision is the bottom-right cell divided by the right column; recall is the bottom-right cell divided by the bottom row.

Figures

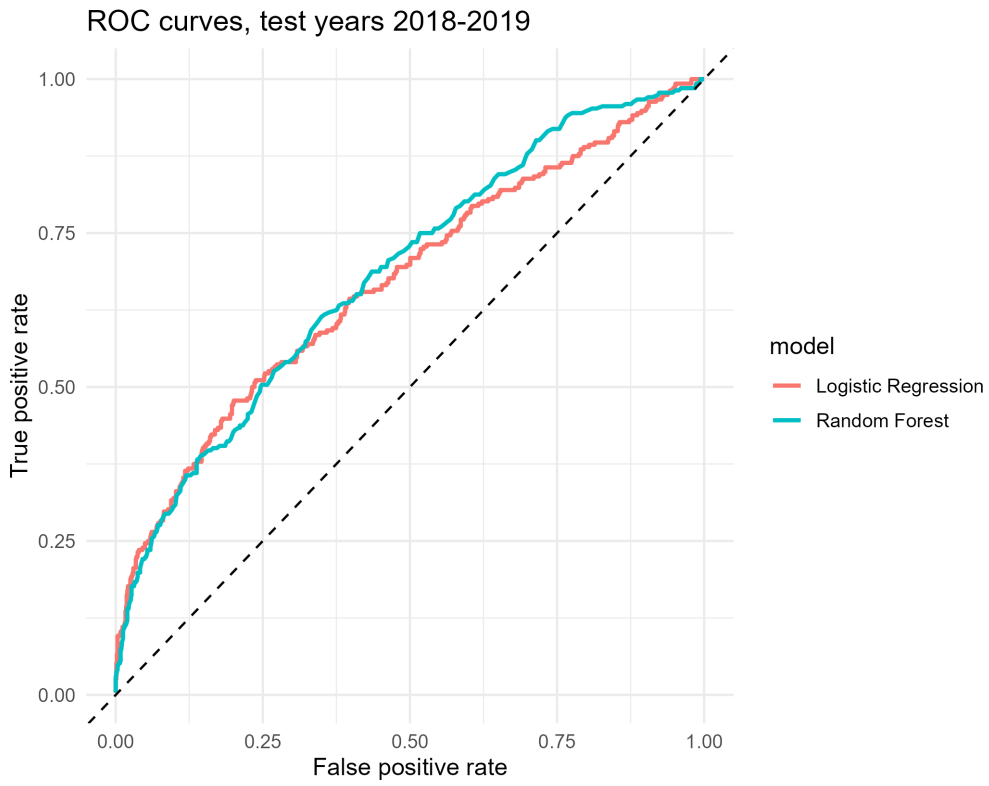


Figure 8: Figure 14: ROC curves, test years

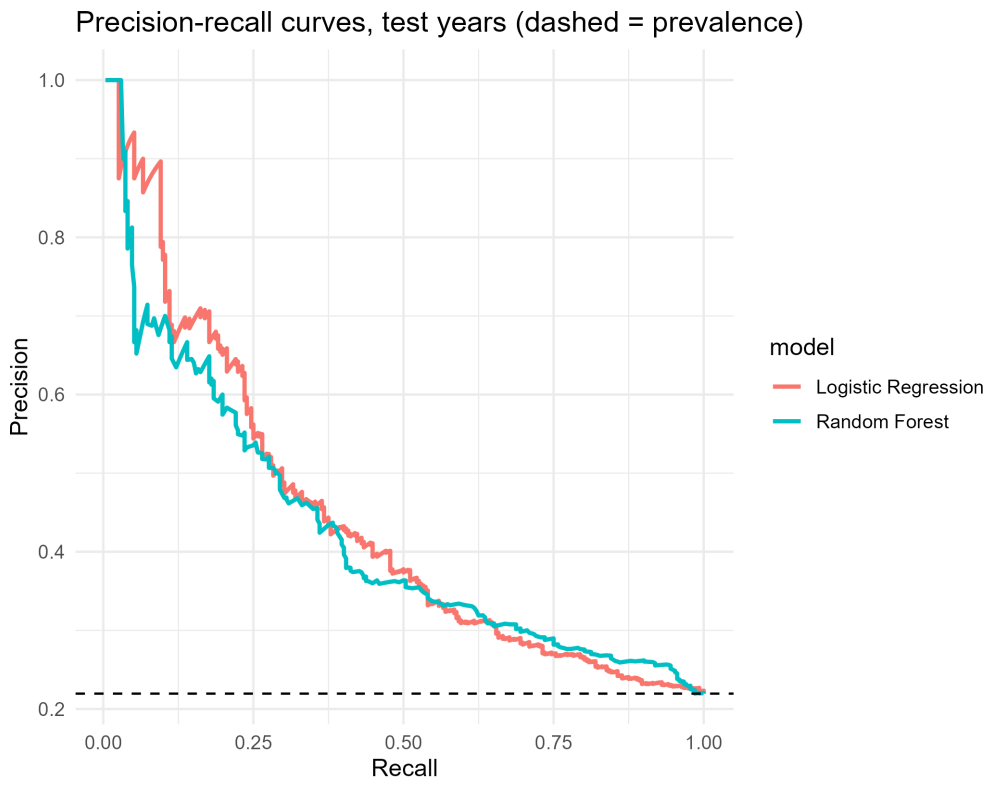


Figure 9: Figure 15: Precision-recall curves, test years

In the ROC curve, a line bowing toward the top-left is better, and the dashed diagonal is guessing. In the precision-recall curve, a line well above the dashed horizontal prevalence line shows real skill. When High outbreaks are a minority, the precision-recall curve is the more honest view.

Figure 16 ranks variables by how much accuracy drops when they are scrambled. In Figure 17, an odds ratio above 1 raises the odds of a High outbreak and below 1 lowers them. An interval that crosses the dashed line at 1 means the effect is not clearly different from none.

Key finding.

- **has_history**: odds ratio 0.56 (95% interval 0.46 to 0.69), meaning 44% lower odds of a High outbreak
- **prev_log_cases**: odds ratio 1.47 (95% interval 1.30 to 1.66), meaning 47% higher odds of a High outbreak
- **DiseaseMalaria**: odds ratio 0.72 (95% interval 0.56 to 0.92), meaning 28% lower odds of a High outbreak
- **zoneSouth**: odds ratio 0.72 (95% interval 0.52 to 1.00), meaning 28% lower odds of a High outbreak
- **DiseaseChikungunya**: odds ratio 1.37 (95% interval 1.07 to 1.74), meaning 37% higher odds of a High outbreak

Each effect holds the other variables fixed. Category effects are relative to the first (reference) category alphabetically.

Which factors matter most?

Individual variable importances can be hard to compare, so the table below adds them up by **factor group** and normalises each forest to 100%. Permutation importance spreads credit between correlated variables, and grouping reduces that distortion.

Factor group	Regression importance (%)	Classification importance (%)
Outbreak history	38.5	37.5
Location	23.6	26.2
Climate	20.2	18.4
Season and month	10.0	10.6
Disease type	7.8	7.3

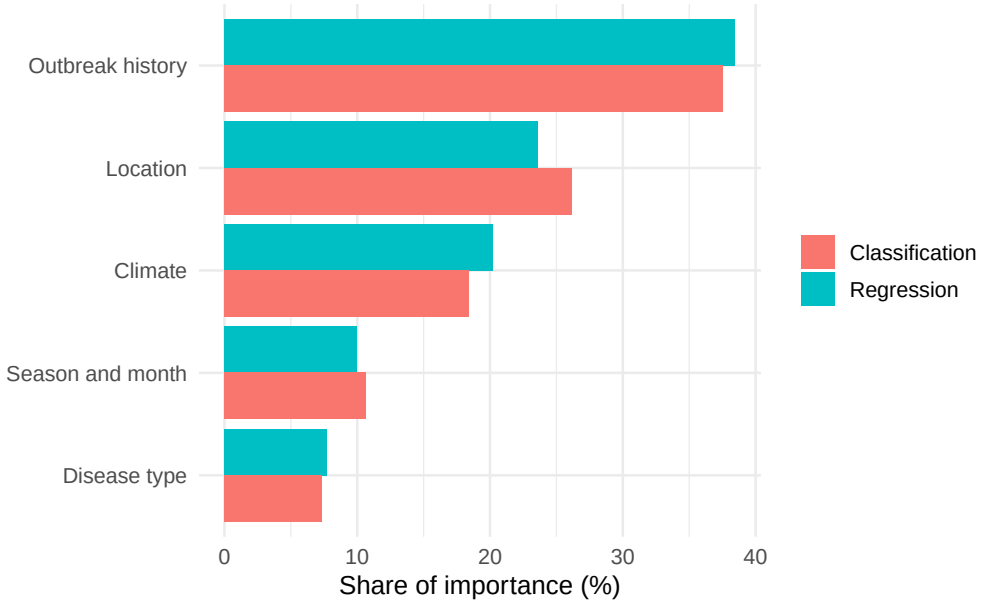


Figure 10: Figure 18: Share of random forest importance by factor group

Key finding.

In the regression forest, **Outbreak history** carries 38% of total importance; in the classification forest, **Outbreak history** carries 38%.

Climate variables together account for **20%** of regression importance and **18%** of classification importance.

Important caution. These scores describe association, not cause. A variable that ranks high helps the model predict outbreak size; it does not prove that changing that variable would change outbreaks.

Clustering outputs

K-Means grouped districts by their outbreak profile (reporting frequency, typical size, share of High outbreaks, disease mix, seasonality and climate). Only districts with at least five reports are profiled.

k	tot_withinss	avg_silhouette
2	4553.382	0.144
3	4064.105	0.146
4	3671.187	0.147
5	3388.765	0.127
6	3144.124	0.143
7	2932.370	0.154
8	2771.287	0.151

In the elbow plot, look for the point where adding clusters stops reducing the within-cluster spread much. The silhouette measures how well each district fits its own cluster compared with the next closest: above 0.5 is strong structure, 0.25 to 0.5 weak, and below 0.25 essentially none.

Key finding.

The analysis uses **7 clusters** with an average silhouette of **0.15**, which indicates **no substantial cluster structure**.

What distinguishes each cluster

cluster	n_districts	total_reports	mean_reports	median_reports	log_cases	share_high	share_medium	share_low	share_dengue	share_chik	share_malaria	temp	precip	lai
1	53	769	14.51	3.01	0.15	0.42	0.36	0.49	0.04	0.03	0.06	30.16	0.32	16.83
2	20	360	18.00	3.41	0.26	0.52	0.27	0.15	0.01	0.01	0.54	28.10	0.41	18.37
3	57	1215	21.32	4.25	0.56	0.68	0.69	0.17	0.02	0.07	0.04	27.15	0.45	13.00
4	60	1781	29.68	3.17	0.13	0.46	0.34	0.27	0.29	0.04	0.03	34.94	0.19	7.48

cluster	n_districts	total_reports	mean_reports	median_reports	mean_log_cases	pct_high	share_monsoon	share_diarrhoeal	share_dengue	share_chik	share_cholera	share_malaria	mean_temp	mean_preci	mean_lai
5	96	1142	11.90	3.07	0.12	0.71	0.87	0.03	0.00	0.02	0.05	29.57	0.45	11.53
6	111	2195	19.77	3.46	0.26	0.47	0.77	0.09	0.06	0.05	0.04	34.45	0.20	7.83
7	46	864	18.78	3.18	0.20	0.58	0.46	0.07	0.03	0.36	0.03	30.25	0.36	13.19

Cluster profiles. SD means standard deviations above (+) or below (-) the average district.

- **Cluster 1** (53 districts) stands out for: share_dengue (+1.6 SD), share_diarrhoeal (-0.9 SD), mean_lai (+0.9 SD).
- **Cluster 2** (20 districts) stands out for: share_malaria (+3.6 SD), share_diarrhoeal (-1.2 SD), mean_lai (+1.1 SD).
- **Cluster 3** (57 districts) stands out for: pct_high (+1.7 SD), median_log_cases (+1.6 SD), mean_temp (-0.8 SD).
- **Cluster 4** (60 districts) stands out for: share_chik (+2.0 SD), share_diarrhoeal (-1.0 SD), mean_preci (-0.8 SD).
- **Cluster 5** (96 districts) stands out for: share_diarrhoeal (+0.9 SD), mean_preci (+0.7 SD), share_monsoon (+0.7 SD).
- **Cluster 6** (111 districts) stands out for: mean_preci (-0.7 SD), mean_temp (+0.6 SD), mean_lai (-0.6 SD).
- **Cluster 7** (46 districts) stands out for: share_cholera (+2.2 SD), share_diarrhoeal (-0.6 SD), share_dengue (-0.4 SD).

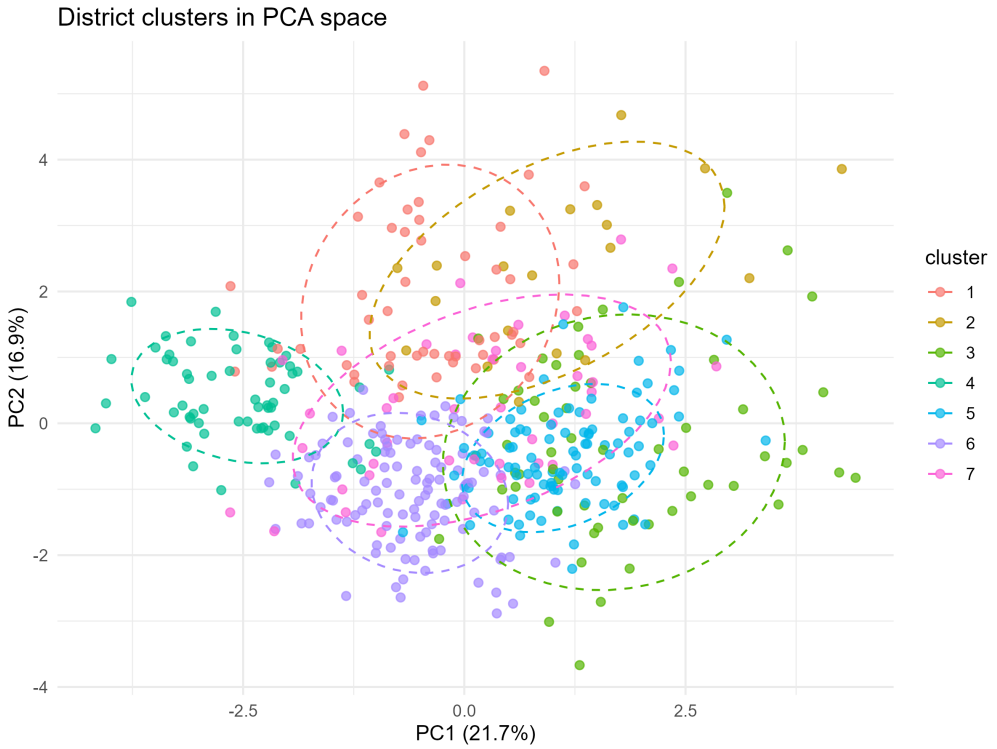


Figure 11: Figure 21: Clusters in principal-component space

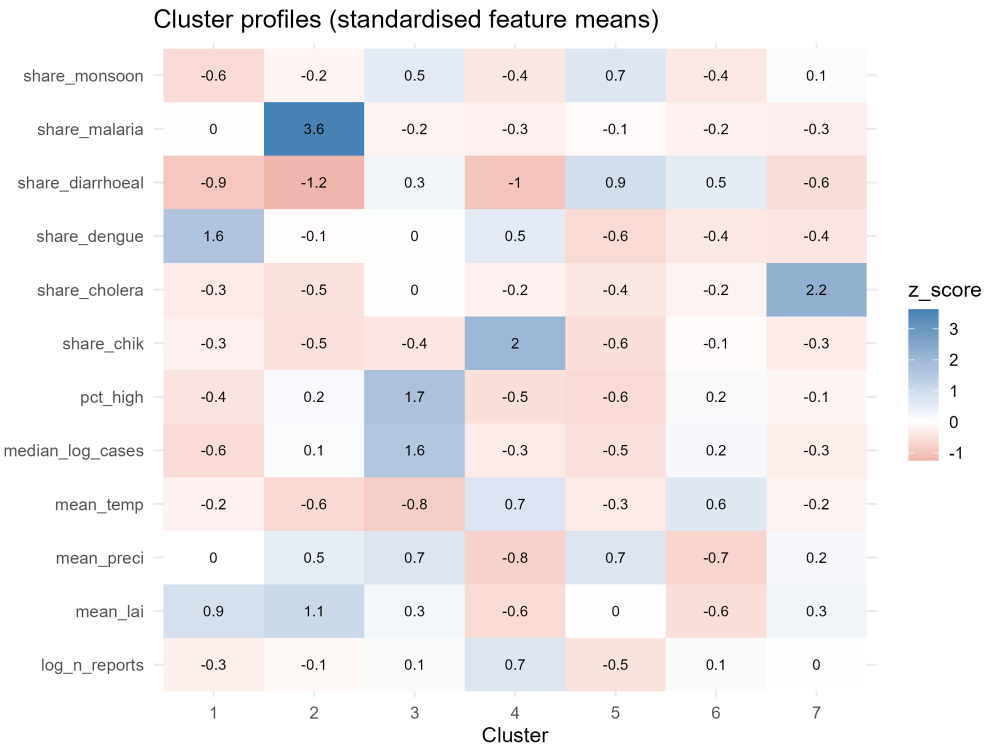


Figure 12: Figure 22: Standardised cluster profiles

Figure 21 flattens the profiles into two dimensions: clusters that overlap heavily are not clearly separated. Figure 22 is a heat map where red or blue cells show which features define each cluster. Figures 23 and 24 show whether clusters are geographic or cut across regions.

Summary of findings by objective

Objective	Headline
1. Disease patterns (time, place, seasonality)	Outbreak sizes are heavily skewed (median 27, mean 89, maximum 51,225 cases).
2. Predict outbreak size (regression)	On test years the best regression model (Random Forest) shows a modest improvement over the baseline (RMSE 0.95 vs 1.05, R-squared 0.18).
3. Predict surges (classification)	On test years the best classifier (Logistic Regression) reaches PR-AUC 0.44 against 0.22 prevalence; it improves on the previous-report rule.
4. Important factors	Most important factor group: Outbreak history (regression) and Outbreak history (classification); climate contributes 20% and 18%.
5. District clusters and rules	Districts form 7 clusters with an average silhouette of 0.15 (no substantial cluster structure).

Conclusion

The project describes outbreak patterns by disease, time and location, tests relationships between recorded factors and case counts, and evaluates models for outbreak-size prediction and surge classification. Clustering and association rules add summaries of district and outbreak patterns.

Model usefulness should be judged against the relevant baseline on held-out test years. Results represent associations in the available records, not causal explanations.

Limitations

This dataset contains reported outbreak events. The analysis therefore focuses on the size and characteristics of recorded outbreaks; it does not directly predict whether a new outbreak will begin. Multiple records can come from the same district, so the usual independence assumptions behind p-values and confidence intervals may not hold fully. Differences in district naming can also split what should be one location into several groups.

Reporting practices may vary across places and years. The 2020–2022 period is especially sparse and unusual, so stress-period results should be treated cautiously. Finally, the patterns identified here are associations. They do not establish that a particular factor causes an outbreak.

Future work

Future improvements could use regular weekly surveillance data, standardise district names, and test spatial or hierarchical models that account for repeated records from the same districts.

References

1. Breiman, L., "Random Forests", *Machine Learning*, 45(1), pp. 5-32, 2001.
2. Rousseeuw, P. J., "Silhouettes: a graphical aid to the interpretation and validation of cluster analysis", *Journal of Computational and Applied Mathematics*, 20, pp. 53-65, 1987.
3. Agrawal, R. and Srikant, R., "Fast algorithms for mining association rules", *Proc. 20th VLDB Conference*, 1994.
4. James, G., Witten, D., Hastie, T. and Tibshirani, R., *An Introduction to Statistical Learning with Applications in R*, 2nd ed., Springer, 2021.